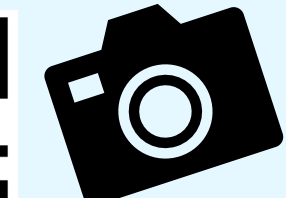


Pathology Foundation Models are Scanner Sensitive

Benchmark and Mitigation with Contrastive ScanGen Loss

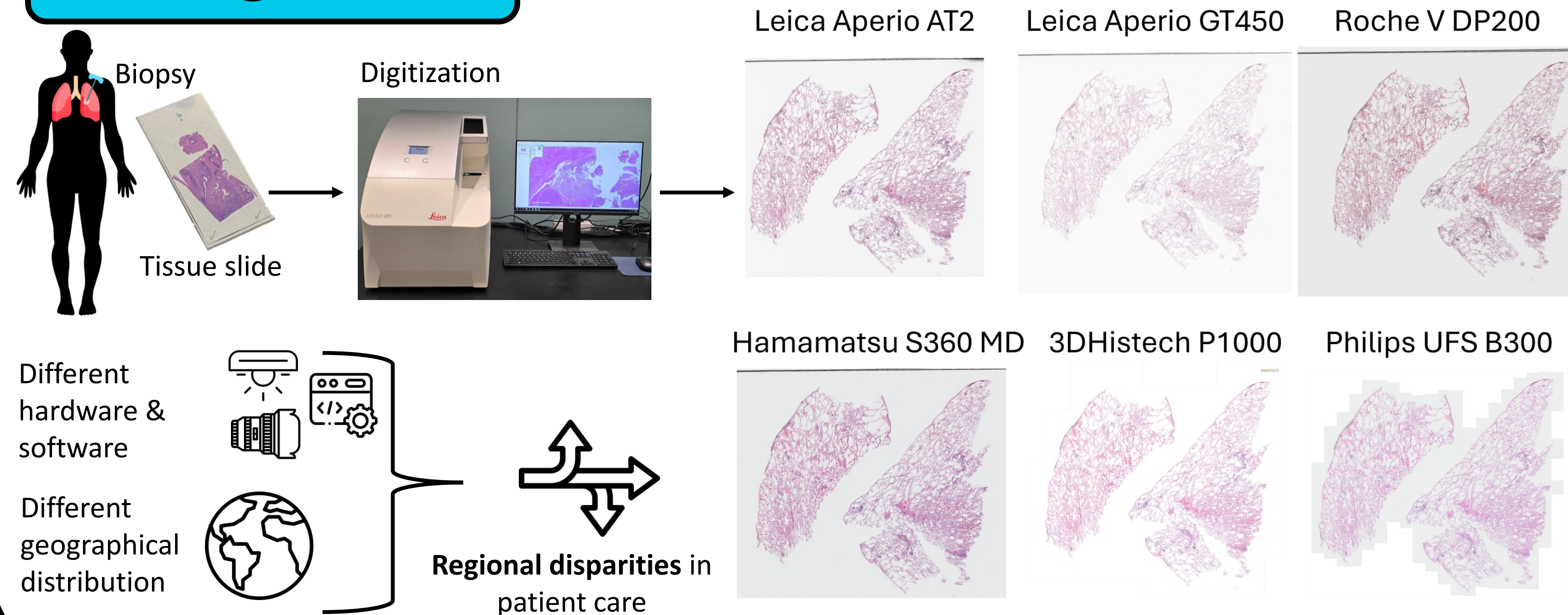
✉ gianluca.carloni@lunit.io *: shared first-authorship

Lunit, Seoul, Republic of Korea



Read our paper!

Background



Contributions

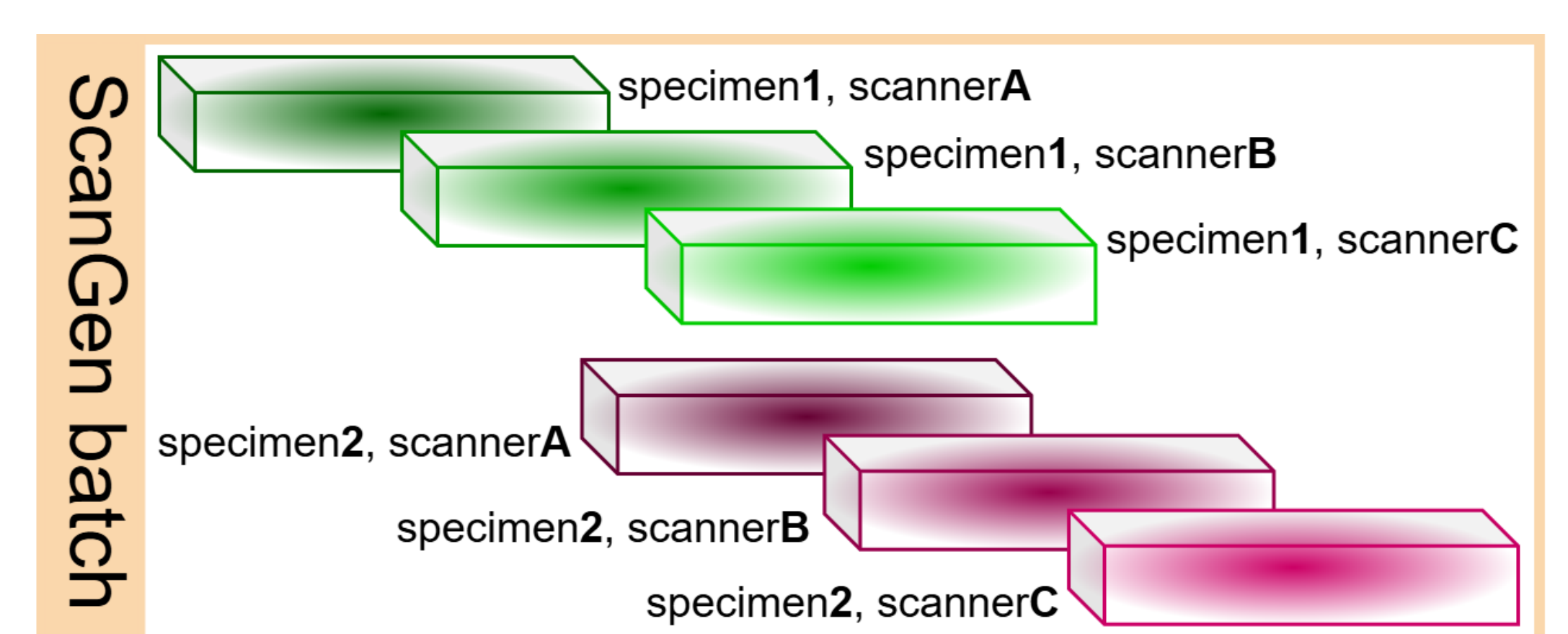
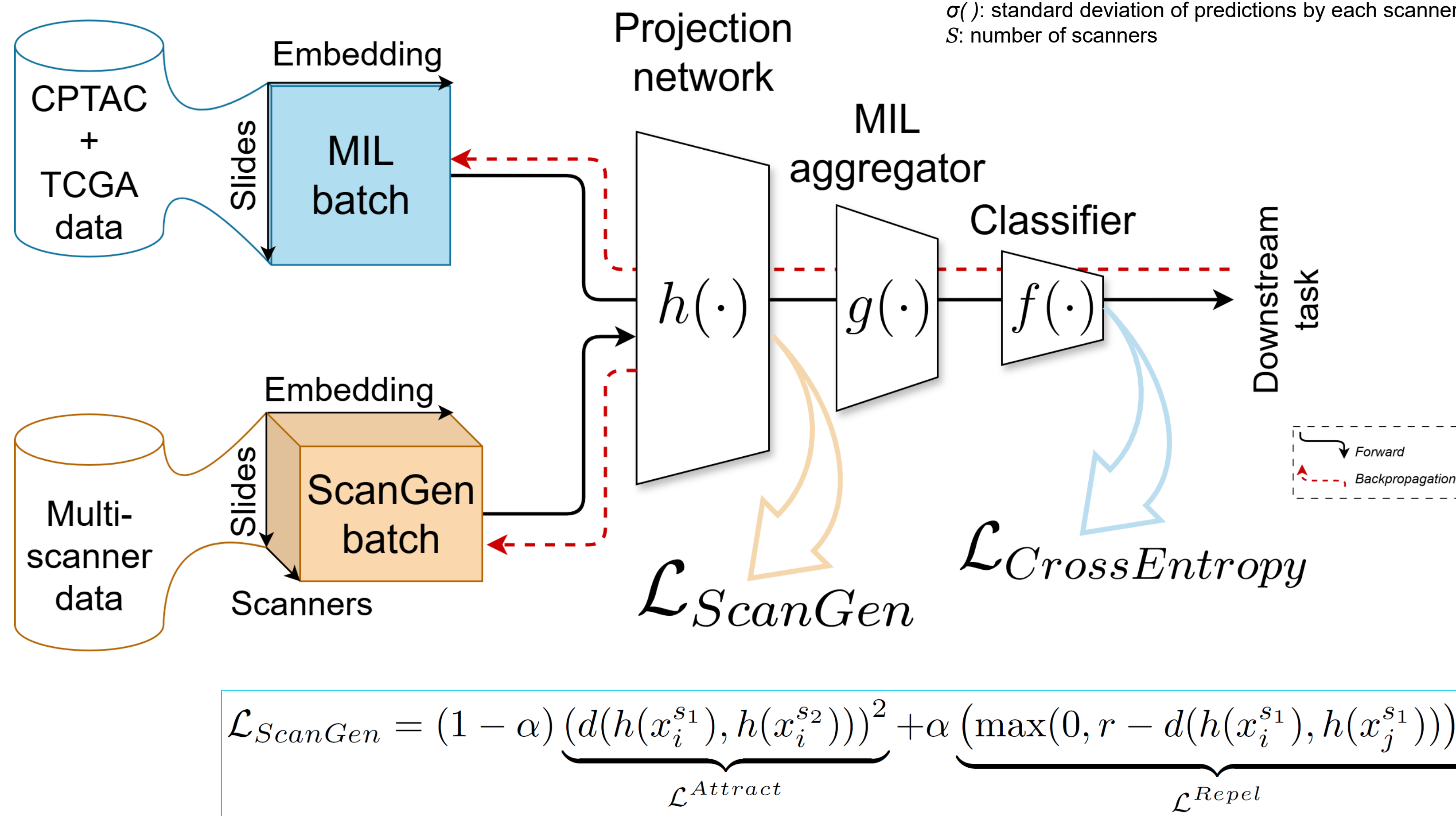
1. Gathered a benchmark of **six** widely used **scanners** and presented evidence of **scanner bias** across cutting-edge **FMs**.
2. Proposed a **contrastive scanner generalization loss** on top of MIL downstream tasks to boost robustness.
3. Applied to **EGFR mutation prediction** in lung cancer, a clinically relevant task.

Methods

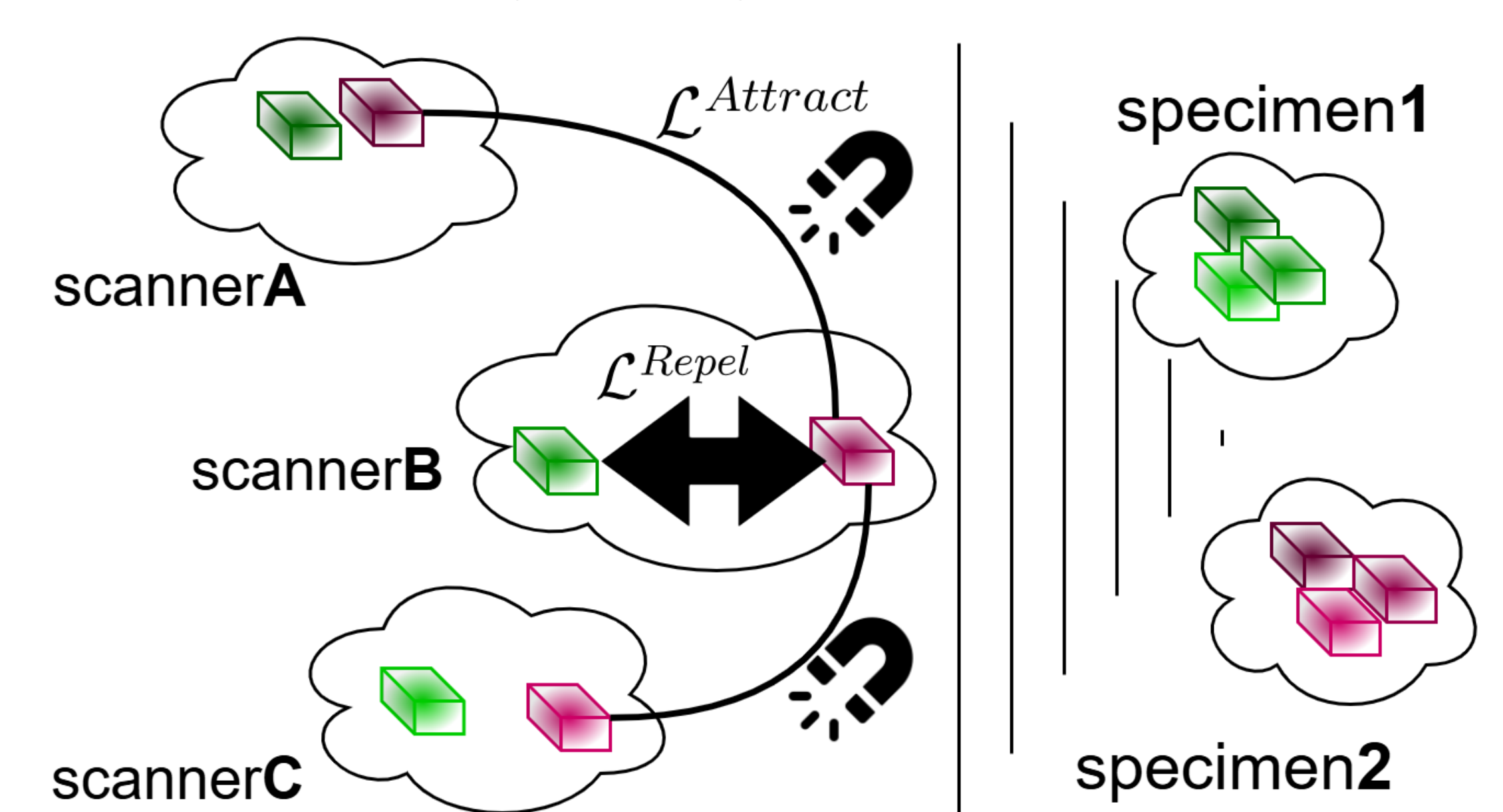
Coefficient of Variation

$$\text{CoV} = \frac{1}{M} \sum_{i=1}^M \left(\frac{\sigma(\hat{y}_i^{s_1}, \hat{y}_i^{s_2}, \dots, \hat{y}_i^{s_S})}{\mu(\hat{y}_i^{s_1}, \hat{y}_i^{s_2}, \dots, \hat{y}_i^{s_S})} \right)$$

M : number of specimens
 $\mu(\cdot)$: mean of predictions by each scanner
 $\sigma(\cdot)$: standard deviation of predictions by each scanner
 S : number of scanners



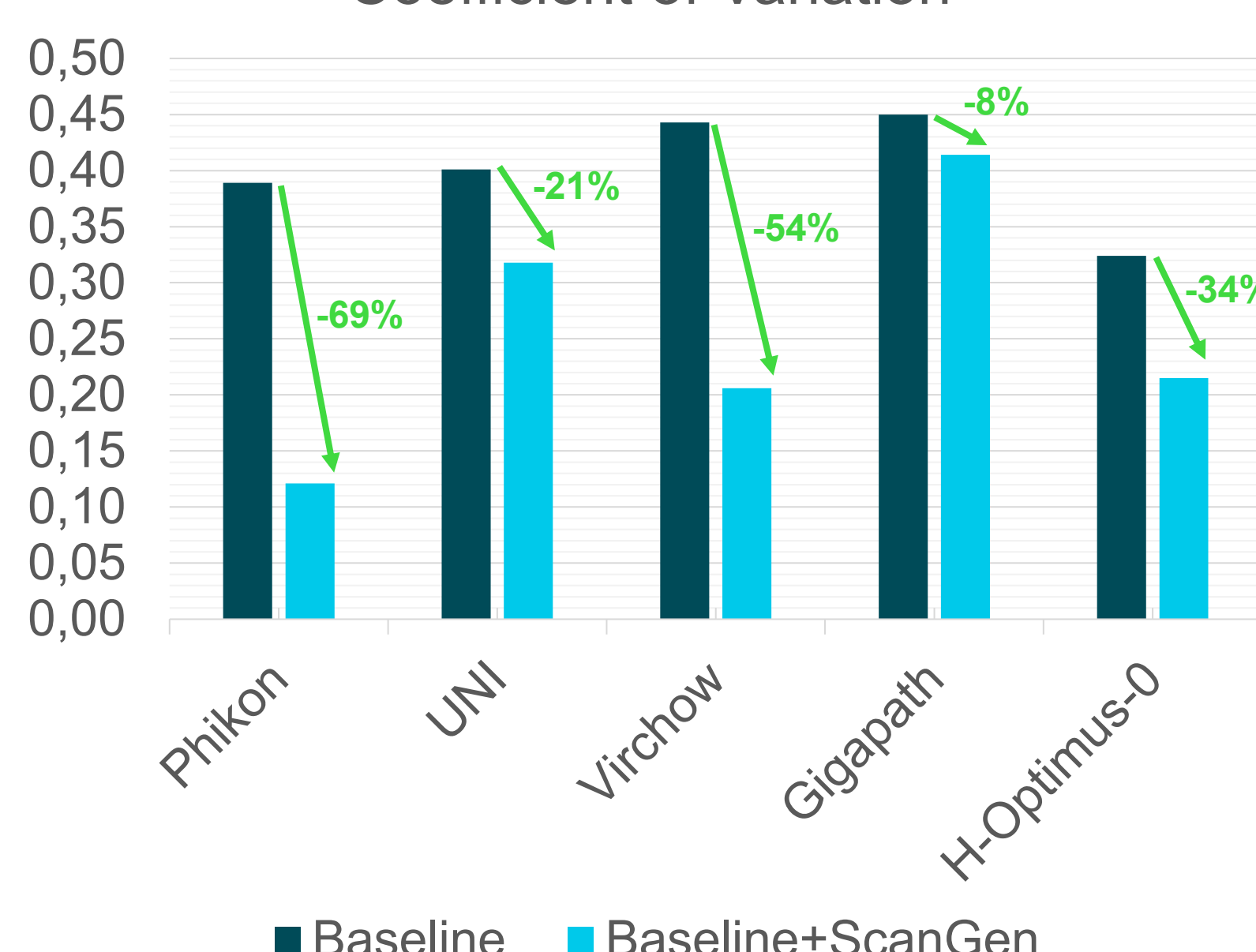
$$\mathcal{L}_{ScanGen} = (1 - \alpha) \mathcal{L}^{Attract} + \alpha \mathcal{L}^{Repel}$$



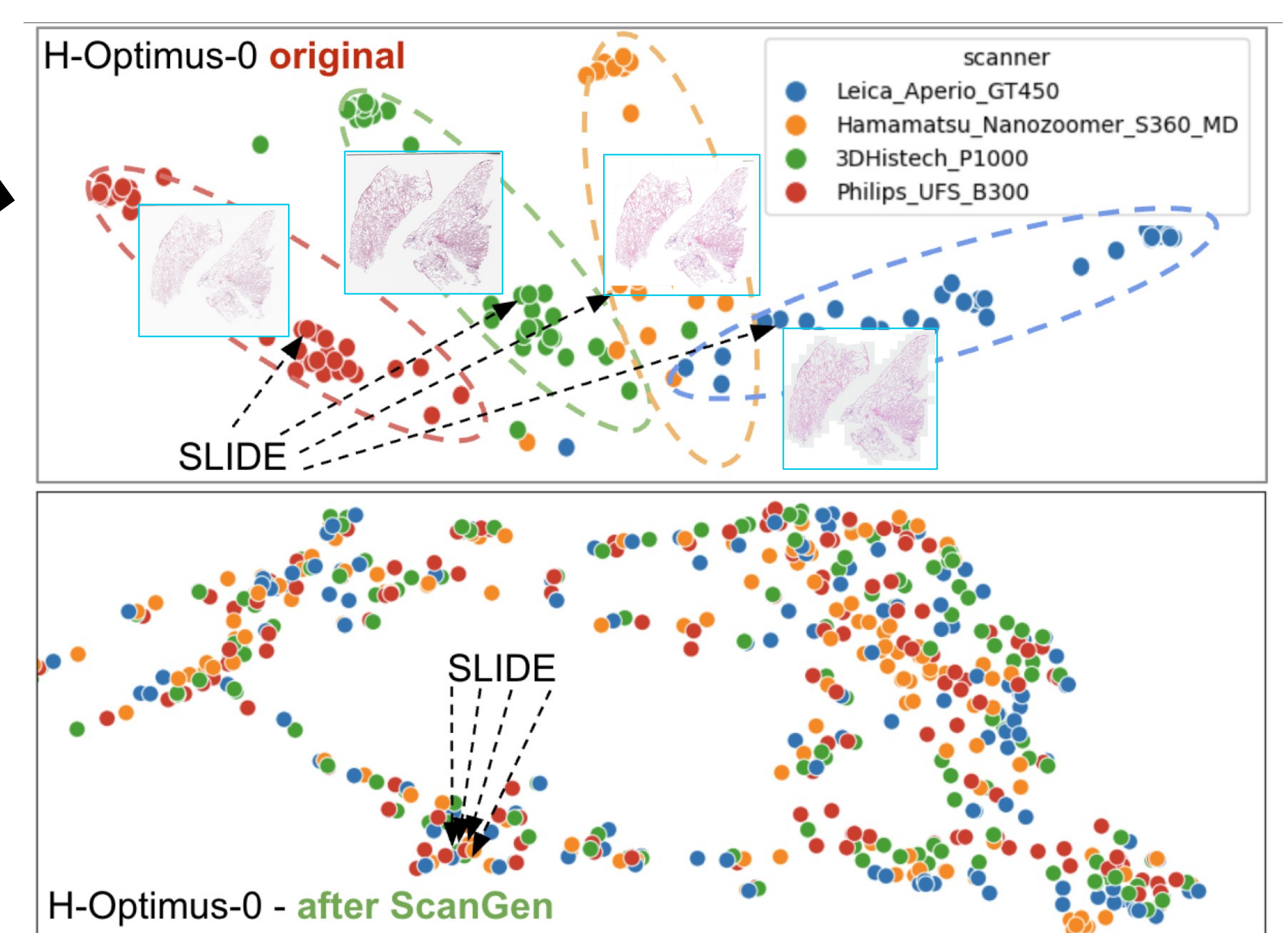
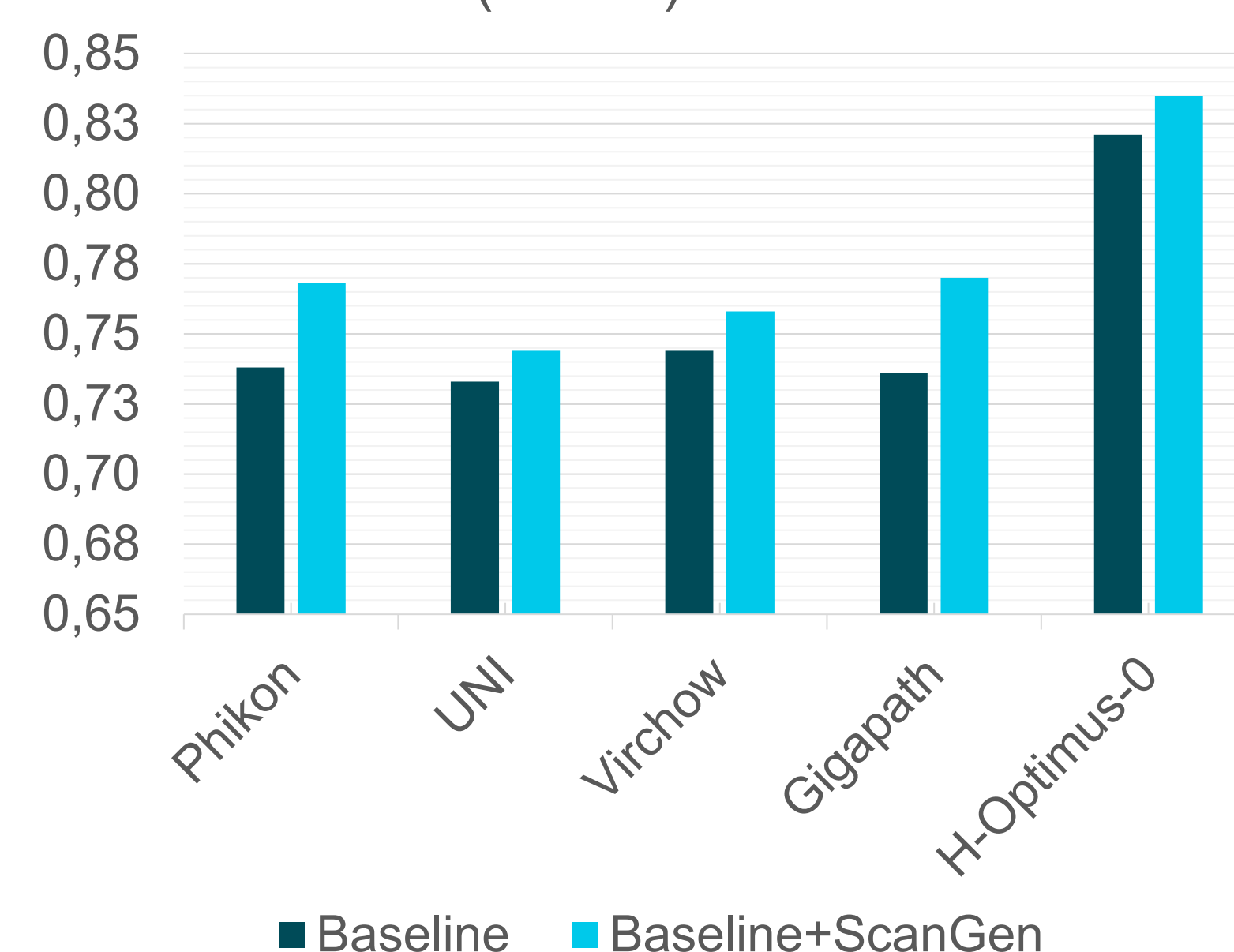
Results

ScanGen **aligns** same-specimen **latents** regardless of scanner

ScanGen Reduces Scanner Coefficient of Variation



ScanGen Improves Downstream Task (EGFR) AUC ROC



ScanGen easily **extends** to more advanced **MIL architectures**

MIL approach	GAP		AB-MIL[8]		DS-MIL[13]		SlotMIL[12]	
	AUC	CoV ↓	AUC	CoV ↓	AUC	CoV ↓	AUC	CoV ↓
EGFR	0.821	0.324	0.828	0.251	0.834	0.265	0.822	0.338
EGFR + ScanGen	0.835	0.215	0.835	0.224	0.828	0.214	0.826	0.227

As low as **three scanners** suffice to **boost** CoV by **~28%**

